

Expert Call Memo — India DNA Sequencer Market, with focus on CE (Capillary Electrophoresis) Sequencers

Expert: Sailesh — former Thermo Fisher Scientific (India); now outside the company

Call title on recording: "Sailesh Clarification"

Our side: Chaitanya, Jatin, Devapriya (names as they appear in the auto-transcript; see Appendix D)

Purpose of call: Clarify the methodology, definitions and specific figures in the DNA sequencer market model (Excel) that Sailesh had shared with us, using a written follow-up questionnaire prepared in advance

Memo prepared from: (1) Follow Up Questionnaire – CE DNA Sequencer (11 questions), (2) Follow Up Questionnaire (2 questions on segment classification and test volumes), (3) call transcript, (4) the Excel data extract (units sold, equipment revenue, reagent revenue, software/parts/other revenue, diagnostic assay kits, and the CAGR sheet)

0. How to read this memo

The memo is organised into ten subject segments. Each of the 13 questions we sent is assigned to exactly one segment, so there is no overlap and nothing is left out. Appendix A is a coverage table listing all 13 questions and where each is dealt with.

Within each segment there are three parts, kept separate on purpose:

- **What the expert said** — his statements only.
- **What the data shows** — figures read off the Excel he shared, checked by us.
- **Our read** — our own interpretation, and where the two do not line up.

Two things to keep in mind throughout:

- **The transcript is incomplete.** The transcription service cut off at 30 minutes with the message "This file is longer than 30 minutes." The recording continues past that point. The last usable line is Devapriya asking "Chetan, do we have any other points to confirm? Maybe regarding the —". Five of our 13 questions were not reached in the transcribed portion. They may well have been answered in the untranscribed remainder, and the full audio should be re-transcribed before this memo is treated as final.
 - **The transcript has no speaker labels** and contains recurring mis-hearings ("Sanofisher" for Thermo Fisher, "cDNA sequencer" for CE DNA sequencer, "short rate" for short read, and so on). Speaker attribution in this memo is inferred from context. Appendix D lists the corrections we applied.
-

1. Data provenance and forecast methodology

Covers Question 1: what is the source / methodology behind the 2023 base data and the CAGRs?

What the expert said

- The whole model is built from **market information he holds personally**, not from any published report and not from any bottom-up survey.
- He **cannot use Thermo Fisher internal reports**, because he has left the company: "internal reports, you know, will be difficult to utilise... because I am now out of [Thermo] Fisher. So it will be basically, majorly market info, which is with me."
- He said the market-information route is **more reliable for CE than for other technologies**, because the CE competitive set is closed and well known: "in CE it is a little much easier because we know that there are no more competitors... we clearly know what kind of numbers are floating... you can also know what kind of revenue even the competitor does. It's a little easy that way."
- He confirmed our reading of the mechanics: a **2023 base year taken from market information, then extrapolated forward to 2035 using a defined CAGR** per line.
- He described the table as "a little extensive one" and said he had used only the market information available to him to fill it.

What the data shows

The sheet works exactly as described: one base-year value for 2023, one constant growth rate per row, compounded through 2035. No year-specific events, no separate price and volume drivers except where noted in Section 6.

There is a mechanical error in the CAGR column. **The printed CAGRs are calculated over 13 years instead of 12 intervals**, so every one is understated by roughly 0.8 percentage points. The underlying assumptions are clean round numbers, which confirms the growth rates were chosen first and the column is simply mis-computed:

Line	2023	2035	CAGR printed in sheet	Actual CAGR in the series
Total DNA sequencers — equipment revenue	27.9	85.9	9.04%	9.83%
NGS short read — equipment revenue	20.0	62.8	9.20%	10.00%
NGS long read — equipment revenue	4.0	10.1	7.36%	8.00%
CE — equipment revenue	3.9	13.1	9.76%	10.62%
Total DNA sequencers — reagents	50.0	383.7	16.97%	18.51%
NGS short read — reagents	40.0	356.6	18.33%	20.00%
NGS long read — reagents	3.0	9.4	9.20%	10.00%
CE — reagents	7.0	17.6	7.36%	8.00%
Total DNA sequencers — software/parts	5.5	25.5	12.52%	13.63%
NGS short read — software/parts	4.0	21.4	13.77%	15.00%
NGS long read — software/parts	0.5	1.6	9.20%	10.00%
CE — software/parts	1.0	2.5	7.36%	8.00%
CE — units sold	52	174	(not shown)	10.59%
NGS short read — units sold	65	204	10 (assumption shown)	10.00%

All revenue figures are USD million.

Our read

- The model is a **single practitioner's estimate**, not a triangulated market model. There is no document we can cite behind the 2023 base year. For anything client-facing this needs to be labelled as expert estimate and, ideally, cross-checked against a second source — at minimum for the NGS lines, where his own confidence is lower (see Section 7).
- His argument that CE is the easier part to estimate is credible and is supported internally by the sheet: the CE section is the only one broken down by manufacturer and model and end use, whereas NGS is a single "total for all manufacturers, total for all models" line.
- **The CAGR column should be corrected before use.** It does not change the forecast values — those were built from the correct round-number rates — but if anyone quotes the CAGR column alongside the 2035 values, the two will not tie.
- The questionnaire's framing of "~10% equipment vs ~8% reagents" for CE is right, and the corrected figures are **10.6% for CE equipment and 8.0% for CE reagents**. That gap is the subject of the next section.

2. CE growth structure: why equipment grows faster than consumables

Covers Question 2: CE equipment CAGR is ~10% while reagents are ~8%. This is the reverse of the usual pattern where consumables are the main driver. Why?

What the expert said

He did not dispute the pattern; he explained it as a **funding-driven instrument market sitting on top of a utilisation-driven consumables market**, with the two disconnected in India:

- **CE is the entry point into sequencing in India.** "Any lab which wants to do sequencing in India — the capillary sequencer is the first thing to jump into."
- **Government funding pushes instruments out regardless of workload.** There are government schemes — he gave thalassemia screening as the example — under which CE systems are bought quickly. "In the government sector in itself, there is a huge funding for capillary sequencer."
- **The eligible government buyer base is wide.** The more AIIMS institutions and IITs there are, the more buyers: he noted the IITs now have biology departments too, and that there are a large number of government medical colleges that will receive CE funding.
- **Those government machines are then underused.** "The unfortunate [thing] is that those machines are not utilised to that extent, unfortunately."
- **Private labs are the reverse.** "Even if private players will buy less, they will utilise it to fullest. And that majorly drives your capillary sequencer reagent business."
- He added, as an aside, that **government is now beginning to receive NGS funding as well** — "slowly they're also getting fund for NGS; earlier they did never used to get" — whereas CE funding has always been available in the government channel.

- On the private side, he noted that a lab adopting genomics buys the whole set together: "they will also buy an NGS, they will also buy a capillary sequencer, they will also buy a qPCR."

What the data shows

The sheet is consistent with this story, and it is worth being explicit about what it implies. CE units grow at 10.6% a year while CE reagent revenue grows at 8.0%. Because units compound faster than consumables, **reagent revenue per instrument has to fall over the forecast**:

- Reagent revenue per unit shipped in the same year: **USD 134,600 in 2023 falling to USD 101,300 in 2035** — a 25% decline.
- Measured against the cumulative installed base rather than annual shipments, the decline is steeper still, because the installed base compounds while consumption does not.

Our read

- The explanation is coherent and specific to India, and it is the single most important structural point to come out of the call. It is not the generic "consumables drive the market" model.
- **But it carries an assumption that should be made explicit in the model:** the share of the CE installed base that is idle keeps rising every year through 2035. The 25% fall in reagent pull per instrument is not stated anywhere in the sheet — it is a by-product of two independently chosen CAGRs. If the client challenges the model, this is where it will be challenged.
- Two follow-ups worth putting to him: (a) roughly what proportion of government-installed CE systems are effectively dormant today, and (b) does he expect that proportion to worsen, hold, or improve as government labs mature? If he expects it to hold rather than worsen, the reagent CAGR is too low relative to the equipment CAGR and the two rates need to be brought closer together.
- Note also that the CE reagent line **excludes assay kits entirely** (see Section 7). Part of the apparent weakness in the 8% reagent growth may be a definitional artefact rather than a real utilisation effect, because the faster-growing diagnostic assay kits are sitting outside this line.

3. CE volume trajectory and the durability of demand to 2035

Covers Question 3: the CE equipment CAGR implies ~174 units sold in 2035. We would expect CE unit sales to flatten or decline, as CE's role in overall sequencing shrinks and testing centres finish building capacity.

What the expert said

He accepted the question as legitimate — "I think this question is very valid, that at what time point will the procurement flatten out" — and then argued that **the flattening does not happen inside the 2025-2035 window**, for five reasons:

- **The government base is far from saturated.** "There are a huge number of medical colleges which do not actually have a capillary sequencer at this juncture."
- **The AIIMS build-out is still running.** Of the 19 AIIMS, "at least 13 to 14 are fully functional, but at least another 10 will become functional within coming years." And each one is not a single sale: "every AIIMS will have at least three to four departments which will require a capillary sequencer — from pathology to microbiology to biochemistry... or immunology. They can all buy a capillary sequencer."
- **Replacement and capacity additions in the private sector.** He conceded that large labs may stop adding CE capacity — "the major labs and all, they might actually cease to procure a CE" — but pointed out this is not the same as stopping purchases. Heavy users wear machines out. His example: MedGenome, whose systems are "maybe just four years old — it will go through wear and tear; if they utilise this heavily, automatically they have to buy new systems as well." He was explicit that private-sector buying covers both drivers: "there will be additional systems which will be procured to manage the load also, and to manage the wear and tear as well, both."
- **Model refresh drives upgrade purchases.** Vendors launch new CE models with advantages that pull customers into replacing working machines: he cited the 3500 being succeeded within Thermo Fisher (transcribed as "autoflex", most likely the SeqStudio Flex family), and said Applied Biosystems, Promega "or somebody else" will keep coming up with new CE models. "They will continue to procure those new systems."
- **Tier-2 and tier-3 city expansion is only just starting.** "It is only in the couple of years that the tier-two and tier-three cities have started their journey in genomics." Beyond the roughly ten well-known tier-2 cities, "there are many more tier-two cities and tier-three cities which are still to adopt." Every new lab opening there buys CE. He noted this is a deliberate corporate strategy across the industry: "if you see tier two, tier three, it is in a solid place... it is there in their strategy."

He was also direct that **NGS does not displace CE**: "It is not going to happen that NGS will completely remove

capillary sequencer. There will always be a scope for capillary sequencer to be adopted."

The one concession he made: "till 2035, I mean, we can bring down the CAGR by one to two percent, but then I think more or less it should stand at that time point... let us stand with that number at this juncture."

He **declined to name a year in which procurement flattens**. When pressed on after how much time the CAGR would drop and by what percentage, he answered by restating that the forecast horizon runs to 2035 and that the market is large, rather than giving a turning point.

Related figure captured on the call

On the government share of clinical-diagnostics CE purchases, he confirmed our earlier number: of roughly **24 CE units bought for clinical diagnostics, about 8 are bought by government — about one third**.

What the data shows

Applying his own concession to the corrected CE growth rates:

Scenario	CE units CAGR	CE units in 2035	Change vs sheet	CE equipment revenue 2035
As modelled	10.6%	174	—	USD 13.1M
Cut 1 pp (his lower bound)	9.6%	156	–10%	USD 11.7M
Cut 2 pp (his upper bound)	8.6%	140	–20%	USD 10.5M

So the concession he offered is worth a 10–20% reduction in the 2035 CE numbers. It is not a rounding adjustment.

Our read

- His reasons split cleanly into two kinds, and they should be treated differently. Reasons 1, 2 and 5 are **new-buyer growth** (unserved government institutions, new AIIMS, new tier-2/3 labs). Reasons 3 and 4 are **replacement demand** from the existing base. Replacement demand is the more durable of the two and is the part our original question underweighted — we were reasoning as if capacity build-out were a one-time event, and he is right that a heavily used CE instrument has a finite life.
- **However, our question is still not fully answered.** New-buyer growth is inherently finite: there is a fixed number of AIIMS, government medical colleges and tier-2/3 cities. A model that compounds at 10.6% for twelve straight years assumes the new-buyer pool never runs down. He asserted the pool is large without sizing it.
- **Recommended action:** rather than a flat CAGR cut, ask him for the size of the addressable base — how many government medical colleges exist, how many currently have a CE, and how many CE units a mature tier-2 lab installs. That would let us build a saturation curve (fast growth to roughly 2030, then replacement-led growth at a lower rate) instead of a straight line. If he will not size it, we should at minimum carry his 1–2 pp cut as a downside case and disclose the 10–20% swing.
- His MedGenome example is the only replacement-cycle data point we have, and it is indicative rather than precise — he described a machine "maybe just four years old" as already showing wear, not as due for replacement at four years. It is worth asking him for the working life of a heavily used CE instrument, because if the figure is around four to five years then a large and rising share of 2030s volume is replacement of machines sold in the 2020s. That would be a defensible reason for volumes not to collapse and should be modelled as a separate replacement line rather than buried in a single growth rate.

4. CE versus NGS short read: similar unit counts, very different revenue

Covers Question 4: CE units sold each year are on par with NGS short read, which contradicts our understanding that the DNA sequencer market is NGS short-read led.

What the expert said

He said there is no contradiction, and that the difference is **ticket size, not unit count**. He worked through it against the sheet on screen:

- NGS short-read **instrument** revenue is "roughly 20, 25 million", while CE is "roughly 5 million only. So there is already a gap... almost a three, fourfold gap."
- On consumables: "the more NGS you utilise, the more expensive are the kits. So that will actually define a huge number of reagents procurement for NGS." He pointed to the NGS short-read reagent line at "almost 57 million, which we are telling is only reagents."
- On the price spread of individual machines: long-read NGS has a **small ticket size** because it is dominated by Oxford Nanopore — "you can get a Nanopore MinION for 20K as well" — with market prices "between 20K to

300K". A high-end short-read NGS system, by contrast, "will be 800K".

- His summary: "the ticket size is higher there. Numbers may be more or less similar."

What the data shows

Average selling price implied by the sheet for 2025 (equipment revenue divided by units sold):

Line	2025 equipment revenue	2025 units	Implied average price per unit
NGS short read	USD 24.2M	79	USD 306,000
NGS long read	USD 4.7M	23	USD 203,000
CE	USD 5.4M	69	USD 78,000

CE and NGS short read do sell in comparable unit volumes (69 vs 79 in 2025), but an NGS short-read system carries about **four times** the price of a CE system, and NGS short-read reagents are roughly **seven times** the CE reagent line (USD 57.6M vs USD 8.2M in 2025). Both effects run the same way, which is why NGS short read dominates revenue while barely leading on units.

Our read

- This resolves the question, and the explanation is arithmetically sound against his own sheet. Our original framing conflated unit share with revenue share; the market is NGS short-read led **on revenue**, and roughly balanced between CE and NGS short read **on units**.
- The CE implied price of USD 78,000 sits at the bottom of the sheet's stated 75–150K band for the 3500 family. That is because the low-priced SeqStudio 4-capillary system (50–60K) makes up a meaningful part of CE volume — 10 of 69 units in 2025. The implied average is internally consistent, not a red flag.
- **Worth stating in any output we produce:** unit counts and revenue tell different stories in this market, so any chart showing "units sold" needs to sit next to one showing revenue, or readers will draw the same wrong conclusion we did.

5. End-use segment economics: forensics versus clinical diagnostics

Covers Question 5: annual sales of ~12 units into Forensics deliver ~20% of CE revenue (including consumables), while ~24 units into Clinical Diagnostics deliver ~60%. Why?

What the expert said

He confirmed the driver is **utilisation intensity per instrument**, and described the two segments concretely:

- **Forensics runs a narrow, low-throughput menu.** "In forensics, that is a very basic requirement, what they run — one or two Identifiler kits, and one or two of those VNTR kits they run."
- **Clinical diagnostics runs the machines hard, continuously.** "The sequencers in the market which are there for clinical, they are used heavily. Every day they run, day in and day out, 24 hours they keep on running — churning out 96, 96, 12 plates. In fact, some of them churn out now four plates per day. So that's a huge requirement."

So a clinical CE instrument consumes several times the consumables of a forensic one, which is why half as many forensic placements produce a third as much revenue.

What the data shows

The revenue-per-unit ratio implied by our question — roughly 20% of revenue from 12 units versus 60% from 24 units — means a clinical unit generates about **1.5 times** the revenue of a forensic unit. His description of the workload difference (continuous 96-well plate running versus a handful of STR kits) suggests the true gap in consumables consumption is wider than 1.5x, with the instrument sale itself diluting the ratio.

One reconciliation problem. The ~12 and ~24 unit figures are not in the extract Sailesh shared with us. His sheet splits CE by model and by "R&D/Forensic" versus "For diagnostic use", and those splits do not produce 12 and 24:

CE end-use split from the sheet	2023 units	2025 units
R&D / Forensic rows	16	22
Diagnostic-use rows	36	46
Total CE	52	69

Two things follow. First, the sheet's "R&D/Forensic" category **combines research and forensics**, so it is not comparable to a standalone Forensics segment. Second, the 12/24 numbers must come from a separate segment-level table (Forensics / Clinical Diagnostics / Research & Academia / Agrigenomics) that we have but he

did not include in this extract — or from an earlier version of the model.

Our read

- The mechanism is confirmed and well evidenced. The specific claim that some clinical sites run four 96-well plates per day is a good, quotable data point on utilisation.
- **The unit numbers need to be reconciled before publication.** As it stands we have two incompatible views of CE end-use volumes: 12 forensics / 24 clinical from the segment table, versus 22 R&D-plus-forensic / 46 diagnostic from this sheet. These need to be tied together, and specifically we need R&D separated from Forensics, because his sheet merges them and our segmentation does not.
- This reconciliation is also a prerequisite for the segment-classification question in Section 10 — if "Medical Research" moves out of Clinical Diagnostics, the 60% revenue share changes as well.

6. Pricing: level, direction and basis

Covers Question 6 (what rate of price increase or decrease should we assume over the forecast) and Question 8 (are the prices list, net-of-discount, or India-landed).

What the expert said

On the basis of the prices (Question 8) — answered clearly. All prices in the sheet are **net of discount**, i.e. realised transaction prices, not list. "These are discounted prices, yes. Whatever we have tried to find out, whatever is the market information, is very clearly from a discounted pricing only, because the list prices are very, very high." He confirmed this is what he expects an institution actually to pay.

On direction (Question 6) he described two opposing forces that largely cancel:

- **Upward: supplier pricing power.** CE is "solely manufactured majorly, or 95 percent of the market, by Applied Biosystems of Thermo Fisher", so the supplier "might actually like to raise the pricing by 5, 10 percent every year".
- **Downward: FX and negotiation.** "Forex and everything will also try to impact it. You would try to increase, but then the forex is also increased, so you might have to negotiate on a lower price as well." The net effect is that realised prices stay inside a band rather than climbing steadily.

He also explained **why the sheet shows wide price ranges for a single machine** — it is buyer mix, not model mix:

- **Government pays more.** "In the government, maybe the prices are a little higher, because they have to manage that with a lot of AMCs, CMCs" — service and maintenance contracts bundled into the capital price.
- **Private pays less.** "In a private, there is a stronger negotiation."
- This is the reason a single 3500 system appears at anywhere from USD 75K to USD 150K. He confirmed this when asked directly whether that range was for the same machine.

What the data shows

The sheet's own price ranges, as shared:

Manufacturer / model	Capacity and use	Price range (net of discount)
Thermo Fisher 3730	96-capillary, large system	USD 250,000-300,000
Thermo Fisher 3500, 3500xL, SeqStudio 8 Flex, SeqStudio 24 Flex	8-24 capillary	USD 75,000-150,000
Thermo Fisher 3500 Dx, 3500xL Dx, SeqStudio 8 Flex Dx, SeqStudio 24 Flex Dx	regulated versions	USD 75,000-150,000
Thermo Fisher SeqStudio	4-capillary, small system	USD 50,000-60,000
Promega Spectrum CE	mid-range	USD 100,000-120,000
Promega Spectrum Compact	4-channel, compact	USD 50,000-70,000
NGS short read (all)	—	USD 40,000-800,000
NGS long read (all)	—	USD 20,000-300,000

There is a direct conflict between what he said and what the model does. For almost every CE row, the unit CAGR and the revenue CAGR are identical, which means **the model assumes a flat average price for the entire forecast** — no increase at all:

CE row	Units CAGR	Revenue CAGR	Implied price trend
3500 family — R&D / forensic	7%	7%	flat
3500 family — diagnostic use	12%	12%	flat

CE row	Units CAGR	Revenue CAGR	Implied price trend
3500 Dx family (non-regulated market)	5%	5%	flat
SeqStudio small — R&D / forensic	5%	5%	flat
SeqStudio small — diagnostic use	10%	10%	flat
Promega Spectrum CE — R&D / forensic	5%	5%	flat
Promega Compact — R&D / forensic	5%	5%	flat
Promega Compact — diagnostic use	8%	8%	flat
3730 — R&D / forensic identification	2.5%	5%	+2.5% a year
3730 — diagnostic use	2.5-5%	2.5-5%	not determinable

Our read

- Question 8 is settled: prices are net-of-discount realised prices. **One part of it was not answered** — whether these are India-landed prices (i.e. inclusive of import duty, freight and local charges) or ex-works equivalent. Given he described them as what an institution pays after negotiation, India-landed is the likely reading, but this needs a one-line confirmation because it materially affects any cost or margin work built on these numbers.
- Question 6 is answered directionally but **the answer and the model disagree, and this needs a decision from us**. He described 5-10% annual list increases substantially offset by FX and negotiation; the model applies a 0% price trend on ten of eleven CE rows and +2.5% on one. Flat is a defensible reading of "the increases get negotiated away", but if that is the intent it should be stated as an explicit assumption rather than left as an accident of two matching CAGRs. As it stands, a reader who takes the 5-10% comment at face value will conclude the model is wrong.
- The government-versus-private price gap is a genuinely useful finding beyond the immediate question: **the same instrument carries a materially higher price in the government channel because AMC/CMC is bundled into the capital cost**. That means government revenue per unit is inflated relative to private, and it should be handled explicitly in any channel-level revenue split. It also partly offsets the utilisation point in Section 2 — government machines may be underused on consumables but they are sold at better instrument prices.
- The 3730 rows are internally inconsistent and should be queried (see Appendix C, flag 8).

7. Scope of "reagents" and "assays", and reconciliation of total market size

Covers Question 7 (which of the diagnostic assays listed are strictly for forensics) plus the definitional and total-market issues raised during the call itself. This was the longest and most consequential part of the discussion.

What the expert said

7.1 The CE reagent line excludes assay kits. Devapriya asked directly whether the CE reagent figure for 2025 — about USD 8.2M — is purely reagent revenue with no assay kits in it. Suresh confirmed: "8.2, 8.6. This is purely the reagent sales without the assay kits... yes, yes, yes, you are correct."

7.2 CE forensics assays are excluded from the model altogether. The sheet carries the note "Forensics assays — 5-8 M not included" against the CE reagent row. So USD 5-8M of CE forensic assay revenue sits nowhere in the model.

7.3 The NGS reagent lines *include* assays — the opposite treatment. "This NGS short read and long read, the reagents — there I have included the reagents which they overall sell, including the running cost as well as the assays." He confirmed this specifically covers the OEM assay portfolios: "majority of the assays which are there for Thermo Fisher is done by Thermo Fisher itself... Illumina also, TSO 500 and everything is included in that number."

7.4 Third-party NGS assays are excluded, and he sized them. "You have some third party, but their numbers would not be greater than 10-15 million." He repeated the bound: "roughly your 10-15 million you can keep aside for assays from third party — will not be more than that."

7.5 Why the two are treated differently. Because we only asked for a diagnostic assay table for CE, he had nowhere to put NGS assays: "in the diagnostic assay part, you only requested for CE. If you would have done NGS here, then I would have brought the assay part of NGS here and not added it there." He also said NGS depth was outside the brief: "the major discussion point was around CE only... NGS, if you want to go in detail into it, I think it was not in the scope, I believe."

7.6 He agreed to fix the definition. When told that the mixed definition makes the reagent line impossible to explain — "for me, summing up would become very difficult, I would not be able to explain what all the components are" — he agreed to either split the figure or annotate it: "if you want to remove that, we can always remove it and write it somewhere in the site, in the comment section — we can write that we have removed 20 million or something for assays, that is okay." Asked for this in writing, he said: "Perfectly, we can do that, we can do that... by the side of it we will write that the assay is not being included, and this is roughly the assay numbers, so that you know that this is purely running reagents." He proposed the wording himself: a note stating that "for reagents, specifically for the NGS, at least for the manufacturers, we have basically added the reagents and assays both."

7.7 The assay kit table column is mislabelled — this one matters. The bottom table is headed "Annual Kit Sales Volume – Million \$". Devapriya asked whether total assay revenue could be obtained by multiplying those figures by the price range and summing, and computed **about USD 104 million**. Sailesh checked his own copy and corrected it: **the column is already revenue in USD million, not volume**. "It is revenue. You know, revenue, 5 million... it is annual kit sales revenue, because it is in million dollars." He accepted the header is wrong: "But yeah, I should have mentioned revenues in itself directly." He confirmed that multiplying it out would wrongly "go to something like 100 million dollar".

7.8 Total market size. He built the 2025 total verbally: NGS instruments "around 25 million", reagents "another 70 million", software "around 7 million", plus third-party assays of 10–15M — "if you add them up, you will reach at roughly 120 million, which we had expected."

7.9 Revenue structure of NGS. Asked how the CE structure (roughly 25% equipment, 60–70% reagents plus assays, under 10% service) looks for NGS, he answered: "around 25 million for instrument, and you will have 10 million for the service or software, and then you minus by 120 — that is the big chunk of assay, done by both the instrument manufacturers as well as third party. So you have around 85 to 90 million." He confirmed this covers short read and long read together: "this 120 million is total for short read and long read both."

7.10 Question 7 (which assays are strictly forensics) was not answered in the transcribed portion. The only related remark is his description of the forensics menu as "one or two Identifier kits and one or two VNTR kits" (Section 5).

What the data shows

Total market reconciliation for 2025. The sheet ties to his verbal USD 120M once third-party NGS assays are added:

Component	2025, USD M
Equipment revenue (NGS short read 24.2 + long read 4.7 + CE 5.4)	34.3
Reagent revenue (NGS short read 57.6 + long read 3.6 + CE 8.2)	69.4
Software, parts and other (NGS short read 5.3 + long read 0.6 + CE 1.2)	7.1
Sheet total	110.7
Plus third-party NGS assays (his estimate, not in the sheet)	10–15
Plus CE forensics assays (sheet note, not in the sheet)	5–8
Adjusted total market	126–134

His verbal "roughly 120 million" corresponds to the sheet total plus third-party NGS assays only (USD 121–126M). It does not include the CE forensics assays, which the sheet note also excludes. **So the defensible 2025 total is USD 126–134M, not USD 120M** — unless the forensics assays are deliberately out of scope, which needs confirming.

CE assay kit table, read correctly as revenue:

Manufacturer	Product	Target disease	Target gene	Annual kit sales revenue, USD M	Price range, USD
Thermo Fisher	MSI	Cancer	Multiple MS	0.1	25
Thermo Fisher	FGA+SMA	Fragile X and SMA	Fragile X and SMA	0.1	25
Thermo Fisher	Identifier	Chimerism, MCC	Multiple STR	0.8	20
Promega	Forensic kit	Chimerism, MCC	Multiple STR	0.8	20
Promega	MSI	Cancer	Multiple MS	0.2	25
MRC Holland	MLPA	Multiple diseases	Multiple loci	1.0	25
Asuragen	FGA+SMA	FGA+SMA	FGA+SMA	0.5	25
Devyser	QF-PCR	Pre-natal testing	Multiple loci	1.0	25
Total				4.5	

The correct total is **USD 4.5M**, and the erroneous multiplied figure is USD 104.5M — a 23-fold overstatement. This confirms both sides of the exchange on the call.

CE revenue structure, 2025, on the corrected basis:

Basis	Equipment	Reagents (+ assays)	Software / service	Total
Sheet lines only	36.6%	55.4%	7.9%	USD 14.7M
Adding the USD 4.5M diagnostic assay kits	28.1%	65.9%	6.1%	USD 19.2M
Also adding forensics assays at USD 6.5M midpoint	21.0%	74.5%	4.5%	USD 25.7M

The middle row is a close match to the structure in our question (~25% equipment, 60–70% reagents plus assays, under 10% service), which is a useful independent check that the CE picture holds together.

NGS revenue structure: his verbal numbers do not tie to the sheet.

Component, 2025	His verbal figure	Sheet (NGS short + long read)	Gap
Instruments	USD 25M	USD 28.9M	–3.9
Software / service	USD 10M	USD 5.9M	+4.1
Reagents plus assays	USD 85–90M	USD 61.2M	+23.8 to +28.8
Total	~USD 120M	USD 96.0M	+24

Our read

- **The definitional asymmetry is the single biggest problem in the model as shared.** Within the same reagent table, the CE row excludes assays while the NGS rows include them. Nobody reading the sheet would know, and the aggregate "Total DNA Sequencers — reagents" line is therefore a mixed-definition sum that should not be quoted as-is. His explanation of how it happened is reasonable, and he has agreed to fix it. **We should not publish the reagent line until he does.**
- **The assay-table mislabelling is caught and closed, and it was worth the call on its own.** Anyone else picking up the sheet would have made the same multiplication Devapriya made and overstated CE assay revenue by USD 100M. The header must be corrected to "Annual Kit Sales Revenue – USD Million" before the file goes anywhere.
- **His verbal NGS build does not reconcile with his own sheet, and the 120M figure is used two ways.** In one place he treats USD 120M as the total DNA sequencer market including CE; in another he says "this 120 million is total for short read and long read both", which would exclude CE. His USD 85–90M for NGS reagents plus assays looks to be derived by residual (120 – 25 instruments – 10 software = 85) rather than read off the sheet, which is why it sits USD 24–29M above the sheet's NGS reagent lines. This gap is too large to leave open: it is roughly a quarter of the total market. **This is the most important item to resolve with him,** and it should be resolved in writing, not verbally.
- **Question 7 remains open,** and the assay table cannot answer it as it stands. The table's own header says it excludes "STR reagents for personal identification", yet Thermo Fisher Identifiler and the Promega Forensic kit are both listed and both are multiplex STR products. Their target disease is given as "Chimerism, MCC" — chimerism monitoring and maternal cell contamination — which are clinical uses, not forensic. So the two products that look forensic by name are recorded as clinical by application. We need him to state, product by product, which kits are forensics-only, which are clinical-only, and which are used in both.
- The USD 5–8M of excluded CE forensics assays should either be brought into the model as a separate line or documented as out of scope. Leaving it in a cell comment is not sufficient — it is roughly the same size as the entire CE reagent line.

8. Supplier attribution within the CE reagent line

Covers Question 9 (is "Company A" in the reagent sales column Nimagen?) and Question 10 (do Promega and Nimagen have equal share in reagent sales?). Both were flagged low priority in our questionnaire.

What the expert said

Neither question was reached in the transcribed portion of the call. Both remain open.

What the data shows

The CE reagent line is broken out across six supplier slots, three of which are anonymised as Company A, B and C:

Supplier	2023, USD M	Share of CE reagents	2035, USD M	CAGR
Thermo Fisher Scientific	5.00	71.4%	12.6	8%
Promega	1.00	14.3%	2.5	8%
Company A	1.00	14.3%	2.5	8%
Company B	0.00	0%	0.0	—
Company C	0.00	0%	0.0	—

Supplier	2023, USD M	Share of CE reagents	2035, USD M	CAGR
Other manufacturers	0.00	0%	0.0	—
Total CE reagents	7.00	100%	17.6	8%

Three observations follow directly:

- **Promega and Company A are modelled as exactly equal** — USD 1.00M each in 2023, identical 8% growth, identical USD 2.5M in 2035. So the answer to Question 10, as the sheet currently stands, is yes: they are assumed to have equal share. Whether that is a researched finding or a placeholder is precisely what we need to establish.
- **The sheet assumes a three-supplier CE reagent market.** Companies B and C and "other manufacturers" are all zero, and the three named slots sum exactly to the CE total. There is no residual.
- **Company A has reagent revenue but no software, parts or service revenue.** In the software/parts table, only Thermo Fisher (0.95) and Promega (0.05) are populated, and they sum to the CE total of 1.00; Company A, B and C are blank.

Separately, his statement that Thermo Fisher/Applied Biosystems holds "95 percent of the market" checks out against the equipment side of his own sheet:

2025 CE equipment	Thermo Fisher	Promega	Thermo Fisher share
Units	64	4	94.1%
Revenue	USD 5.3M	USD 0.3M	94.6%

Our read

- Observation 3 supports the hypothesis that **Company A is a reagent-only supplier with no instrument installed base** — which is consistent with Nimagen, a third-party Sanger reagent supplier that does not sell CE instruments. This is a reasonable inference but it is not confirmation, and we should get him to name it rather than infer it.
- The exact equality of Promega and Company A at USD 1.00M each, growing at exactly the same rate, reads as a **placeholder split rather than a researched estimate**. Two suppliers in a real market rarely hold identical share to three significant figures and then grow in lockstep for twelve years. This should be treated as unverified until he confirms otherwise.
- **The anonymised labels have to be resolved before anything is published.** "Company A" cannot appear in a client deliverable, and if the name cannot be confirmed the line should be relabelled as "other third-party reagent suppliers".
- Note the contrast between the reagent market and the instrument market: Thermo Fisher holds roughly 95% of CE instruments but only 71% of CE reagents. That gap is the third-party reagent opportunity, and it is worth calling out explicitly — it is a more interesting finding than either of the two questions we actually asked.
- The 95% Thermo Fisher claim is **verified** against the sheet and can be used with confidence.

9. Software, parts, service and other revenue

Covers Question 11: why does short-read NGS increase its share to 84% of software, parts and other sales revenue? Is this driven mainly by installed-base growth, higher software spending, service contracts, data storage or replacement parts?

What the expert said

This question was not reached in the transcribed portion of the call. It remains open.

What the data shows

The sheet does, however, define what this line contains, and the definitions differ by technology:

- **NGS short read and long read:** "Only the maintenance for NGS; softwares FOC from OEM; 3rd party reporting softwares not included." So this line is **maintenance and service only** — OEM software is given away free of charge and third-party reporting software is excluded.
- **CE:** "Software and maintenance included." So for CE, software is inside the line.

The rise to 84% is a mechanical result of the growth assumptions, not an independently estimated share:

Year	NGS short read, USD M	Total, USD M	Short-read share
2023	4.00	5.50	72.7%
2025	5.29	7.06	74.9%
2030	10.64	13.33	79.8%

Year	NGS short read, USD M	Total, USD M	Short-read share
2035	21.40	25.49	84.0%

NGS short read is assumed to grow at 15% a year while CE grows at 8% and NGS long read at 10%. Compounded over twelve years, the faster-growing line necessarily takes share. Nothing else is driving the number.

Our read

- The 84% is an **output of the CAGR assumptions, not an input**. The real question to put to him is therefore not "why 84%?" but "why is NGS short-read maintenance revenue assumed to grow at 15% a year — faster than NGS short-read instruments at 10%?" That is the assumption that produces the result.
- **Our list of candidate drivers can be partly narrowed using the sheet's own definitions**. Because OEM software is free of charge and third-party software is excluded, higher software spending cannot be a driver for the NGS lines. Data storage is not mentioned anywhere and appears to be outside scope. That leaves only two live drivers inside the stated definition: **growth in the installed base under contract, and the price or attach rate of service contracts**. He should be asked to confirm which of the two, and in what proportion.
- There is a plausible reason for maintenance to outgrow instruments — as an installed base matures, a larger share of it comes off warranty and onto paid service contracts, so service revenue compounds faster than new placements for a period. If that is his reasoning it is defensible and should be written down. If not, the 15% needs revisiting.
- **Flag the inconsistent definition across the same table**, as with reagents in Section 7: CE includes software, NGS does not. The "Total DNA Sequencers" line in this table is again a mixed-definition sum.
- Also note his verbal figure of USD 10M for NGS software and service in 2025 against the sheet's USD 5.9M (Section 7). The same USD 4M discrepancy sits inside this table and should be resolved at the same time.

10. Segment classification and test-volume flow

Covers the second follow-up questionnaire: Question F1 (which segment does "Medical Research" belong to, and the percentage splits by segment) and Question F2 (tests originated versus tests performed, by provider type).

What the expert said

Neither question was reached in the transcribed portion of the call. Both remain open. Given the transcript cut off at 30 minutes and the last audible line is Devapriya starting a new question, it is possible these were covered in the untranscribed remainder — the full audio should be checked before re-sending them.

What we asked, and why it matters

F1 — classification of "Medical Research". We need to know whether "Medical Research" sits under Clinical Diagnostics or under Research & Academia. Our reason: internationally, Clinical Diagnostics means tests performed in hospitals for clinical practice and revenue-generating diagnostic services, while Medical Research is normally classified under Research & Academia. If Medical Research is being counted inside Clinical Diagnostics in this model, then **the 55-60% of the market shown as Clinical Diagnostics is unusually high compared with other countries** — which would be a presentational and credibility problem, not just a labelling one. We asked him to fill in this table:

Segment	Subsegments (% split)
Clinical Diagnostics	Patient diagnostics (x%), Medical research (y%)
Research & Academia	
Forensics	
Agrigenomics	

F2 — tests originated versus tests performed. He has already given us tests performed. We asked for tests originated by the same provider types, so we can see how much testing is outsourced and confirm that the originated volumes add up to the performed volumes. The data he has already provided:

Provider type	Tests performed (share)	Tests performed (volume)
Specialised genomics labs	~40%	~200K-240K
National chain labs	~20%	~100K-120K
Regional chain labs	Negligible	Negligible
Standalone labs	Negligible	Negligible
Specialty treatment hospitals	~10%	~50K-60K
Large private hospitals	~10-15%	~50K-90K

Provider type	Tests performed (share)	Tests performed (volume)
Small and medium private hospitals	Negligible	Negligible
Government tertiary hospitals / public medical colleges / programme-linked centres	~15-20%	~75K-120K
Total CE-relevant genetic tests in India		~500K-600K

What the data shows

The tests-performed data he has already given is internally consistent, which is worth noting:

- The percentages sum to **95-105%** across the range, centring on 100%.
- The absolute volumes sum to **475K-630K**, against his stated India total of 500K-600K.

So there is no arithmetic problem with what he has provided. What is missing is the originated-versus-performed split.

Our read

- **F1 is the higher-priority of the two**, because it can move a headline number. If Medical Research is currently inside Clinical Diagnostics and is reclassified out, the Clinical Diagnostics share falls from 55-60% to something closer to international norms, and the Research & Academia share rises correspondingly. That in turn changes the ~60% clinical diagnostics revenue share discussed in Section 5, and it changes the government-versus-private read, because a large part of Indian medical research sits in government institutions. Note that this connects directly to his Section 2 argument: much of what he described as government CE demand — AIIMS departments in pathology, microbiology, biochemistry and immunology — sounds closer to medical research and clinical service combined than to pure revenue-generating diagnostics. **This may be the same issue seen from two directions, and resolving F1 may also explain the low utilisation of government machines.**
- **F2 matters for a different reason.** The concentration of testing in specialised genomics labs (~40%) alongside "negligible" volumes for regional, standalone and small hospitals implies heavy outsourcing — small providers originate tests and send them to specialised labs. If confirmed, that has a direct read-through to instrument placement: a lab that outsources does not buy a CE system. This bears on the tier-2/tier-3 growth story in Section 3, because it raises the question of whether new tier-2/3 labs will buy their own instruments or simply act as collection points feeding the specialised labs. **That is a genuine risk to his CE unit forecast, and it should be put to him directly.**
- Both questions should go back to him in writing along with the items in Section 11.

11. Commitments the expert made on the call

These are the things he agreed to do. They are all definitional or presentational fixes to the sheet, and the first three he committed to without reservation.

#	Commitment	His words	Status
1	Provide a written split of NGS reagent revenue into running reagents versus assays	"Perfectly, we can do that, we can do that"	Agreed — to be received
2	Add a note in the sheet stating that for NGS, OEM reagents and assays are combined, and give the approximate assay amount included	"By the side of it we will write that the assay is not being included, and this is roughly the assay numbers, so that you know that this is purely running reagents"	Agreed — to be received
3	Correct the assay table column header from "Annual Kit Sales Volume - Million \$" to annual kit sales revenue	"I should have mentioned revenues in itself directly"	Acknowledged — needs doing
4	Reduce the CE CAGR by 1-2 percentage points if we want a more conservative view	"We can bring down the CAGR by one to two percent, but... let us stand with that number at this juncture"	Offered, but he preferred to keep the current number. Our decision to make

On item 4, note that his preference is to leave the forecast as-is. If we take up the offer, the 2035 CE numbers fall 10-20% (Section 3).

12. Open items to go back to him with

Ordered by how much they affect the numbers.

Priority	Item	Why it matters
1	Reconcile his verbal NGS figures with his own sheet: USD 85–90M for NGS reagents plus assays versus USD 61.2M in the sheet, and USD 10M for software/service versus USD 5.9M	A gap of roughly USD 24–29M, about a quarter of the total market. Until this is closed we cannot state a total market size with confidence
1	Confirm whether the total 2025 market is USD 120M or USD 126–134M, i.e. whether CE forensics assays (USD 5–8M) are in scope, and confirm whether "USD 120M" includes CE or is NGS-only	He used the 120M figure both ways on the call
2	Deliver commitments 1–3 in Section 11 in writing	The reagent line cannot be published until the definition is fixed
2	Question 7: state product by product which of the eight listed assay kits are forensics-only, clinical-only, or both — specifically Identifiler and the Promega Forensic kit, which are STR products recorded against clinical applications	The assay table currently contradicts its own header. Needed for any segment-level assay split
2	F1: does "Medical Research" sit in Clinical Diagnostics or Research & Academia, and what are the four segment splits	Could move the Clinical Diagnostics share off 55–60%, which affects Section 5
2	Reconcile the ~12 forensics / ~24 clinical diagnostics unit split against the sheet's 22 R&D-plus-forensic / 46 diagnostic split, and separate R&D from Forensics	Two incompatible views of CE end-use volumes are currently in circulation
3	Size the addressable base for CE: how many government medical colleges exist, how many already have a CE, how many CE units a mature tier-2/3 lab installs	Would let us replace the flat 10.6% CAGR with a saturation curve, and answer the flattening question properly
3	Ask for the working life of a heavily used CE instrument. He noted wear at about four years but gave no replacement interval	Would let us split the forecast into new placements and replacement demand instead of one blended growth rate
3	Confirm the intended price trend: the model holds average prices flat while he described 5–10% list increases offset by FX and negotiation	Needs to be an explicit stated assumption, not an accident of matching CAGRs
3	F2: tests originated by provider type	Tests the outsourcing risk to the tier-2/3 instrument growth story
3	Question 9 and 10: is Company A Nimagen, and is the equal Promega / Company A split researched or a placeholder	"Company A" cannot go into a deliverable, and the identical split looks like a placeholder
3	Question 11: why is NGS short-read maintenance assumed to grow at 15% a year, faster than NGS instruments at 10%	The 84% share is an output of this assumption
4	Confirm whether prices are India-landed (inclusive of duty and freight) or ex-works equivalent	Part of Question 8 that was not explicitly answered
4	Query the 3730 rows, where the stated CAGR does not reproduce the printed series	Small in value; a data-integrity fix
4	Fill in the "Approval status by Indian medical device authorities" column, which is blank for all eight assay kits	Relevant to regulated-market sizing

Appendix A — Coverage of all 13 questions

Every question is assigned to exactly one segment.

#	Question (abbreviated)	Segment	Answered?
1	Source / methodology for 2023 data and the CAGRs	1	Yes — personal market information; 2023 base extrapolated at fixed CAGR; no internal Thermo Fisher reports available
2	CE equipment CAGR ~10% vs reagents ~8%, contrary to the usual pattern	2	Yes — government funding pushes instruments; those machines are underutilised; private utilisation drives reagents

#	Question (abbreviated)	Segment	Answered?
3	~174 CE units in 2035; sales should flatten or decline	3	Yes, partly — five reasons why it does not flatten before 2035; offered a 1-2 pp CAGR cut; declined to name a flattening year
4	CE units on par with NGS short read, contrary to an NGS-led market	4	Yes — ticket size, not unit count: instrument revenue USD 5M vs 20-25M, plus far more expensive NGS kits
5	~12 forensics units → ~20% of revenue vs ~24 clinical units → ~60%	5	Yes — utilisation intensity: forensics runs one or two STR kits; clinical runs 24 hours a day, up to four plates daily
6	What rate of price increase or decrease to assume	6	Yes — 5-10% attempted increases offset by FX and negotiation; government pays more (AMC/CMC bundled), private negotiates harder. Conflicts with the model's flat prices
7	Which of the listed diagnostic assays are strictly forensics	7	No — not reached in the transcribed portion
8	Are prices list, net-of-discount, or India-landed	6	Yes, partly — net of discount, confirmed. India-landed not explicitly confirmed
9	Is Company A in the reagent sales column Nimagen? (low priority)	8	No — not reached
10	Do Promega and Nimagen have equal share in reagent sales? (low priority)	8	No — not reached. The sheet does model them as exactly equal
11	Why does short-read NGS reach 84% of software, parts and other revenue	9	No — not reached. It is a mechanical output of a 15% growth assumption
F1	Does "Medical Research" belong to Clinical Diagnostics or Research & Academia; segment splits	10	No — not reached
F2	Tests originated vs tests performed, by provider type	10	No — not reached

Summary: 6 of 13 answered, 2 partly answered, 5 not reached. All five unreached questions fall after the 30-minute transcription cut-off, so the full audio should be re-transcribed before re-sending them.

Appendix B — Additional facts captured, beyond the questionnaire

Points he made that we did not ask for and that are worth keeping.

Market structure

- Thermo Fisher / Applied Biosystems holds about 95% of the Indian CE market. Verified against his sheet at 94.1% of units and 94.6% of equipment revenue for 2025.
- Thermo Fisher holds a much lower share of CE reagents — 71% in the sheet — so roughly 29% of the CE reagent line is third party.
- Promega sells only fragment analysis on the Spectrum CE system, not Sanger sequencing: the sheet notes "no Sanger seq, only fragment" and "only fragment - no marketing" against the Promega diagnostic rows, which is why Promega's diagnostic-use Spectrum CE units are modelled at zero throughout.
- The SeqStudio 4-capillary system is "not promoted by TMO" and is used by "only standalone lab / 2, 3 tier labs" per the sheet notes.
- Promega volumes are modelled low with the caveat "unless Promega focus on aggressive marketing" — i.e. the Promega forecast is explicitly conditional on a commercial decision Promega has not yet made.

Demand drivers

- Government schemes are a named driver, with thalassemia screening given as the example.
- Of the 19 AIIMS, 13-14 are fully functional and about 10 more will become functional in coming years. Each has three to four departments that can each buy a CE — pathology, microbiology, biochemistry, immunology.
- IITs are now CE buyers too, having added biology departments.
- Roughly one third of clinical-diagnostics CE units are bought by government (about 8 of 24).
- A private lab adopting genomics typically buys NGS, CE and qPCR together, not one of them.
- Tier-2 and tier-3 cities began adopting genomics only in the last couple of years, and are named in corporate strategy across the industry.

Utilisation and equipment life

- Clinical CE systems run 24 hours a day; some sites now run four 96-well plates per day.

- Forensic CE systems run a narrow menu — one or two Identifiler kits and one or two VNTR kits.
- Heavy private use produces visible wear within about four years — MedGenome cited as the example, with systems "maybe just four years old" already subject to wear and tear — which generates replacement demand. He did not state a working life figure.
- Model refresh also drives replacement: the 3500 has been succeeded in the Thermo Fisher line (transcribed as "autoflex", most likely SeqStudio Flex).

Pricing and technology

- Oxford Nanopore MinION is available from about USD 20,000; long-read systems range USD 20,000–300,000; a high-end short-read NGS system runs to about USD 800,000.
- List prices are "very, very high" relative to the discounted prices in the sheet.
- Government instrument prices are higher because AMC/CMC service obligations are bundled into the capital price.

Scope statements

- He considered NGS outside the brief: "the major discussion point was around CE only... NGS, if you want to go in detail into it, I think it was not in the scope."
- He is confident on CE and explicitly less so on NGS detail — relevant to how much weight we place on each part of the model.

Appendix C — Data quality flags from our own review of the sheet

These are issues we found in the file, separate from what was discussed on the call. Flags 1 to 5 need fixing before anything is published.

- **The CAGR column is computed over 13 years instead of 12 intervals**, understating every printed CAGR by about 0.8 percentage points. The forecast values themselves are correct. Table in Section 1.
 - **The assay table column headed "Annual Kit Sales Volume - Million \$" actually contains revenue**. Reading it as volume and multiplying by price gives USD 104.5M instead of the correct USD 4.5M. Confirmed by the expert on the call.
 - **"Reagent Sales" is defined inconsistently within the same table**. The CE row excludes assay kits; the NGS rows include OEM assays. The "Total DNA Sequencers" reagent line is therefore a mixed-definition sum and should not be quoted until fixed.
 - **"Software, Parts and Other Sales" is also defined inconsistently**. For NGS it is maintenance only, with OEM software free of charge and third-party reporting software excluded. For CE, software is included. The total line is again a mixed-definition sum.
 - **USD 5-8M of CE forensics assay revenue is excluded from the model entirely**, per the sheet's own note. That is roughly the same size as the whole CE reagent line (USD 8.2M in 2025), so it cannot be left in a cell comment.
 - **The assay table contradicts its own header**. The header says it excludes "STR reagents for personal identification", yet Thermo Fisher Identifiler and the Promega Forensic kit are listed and both are multiplex STR products. Both are also tagged to "Chimerism, MCC" — clinical applications, not forensic.
 - **The "Approval status by Indian medical device authorities" column is blank for all eight assay kits**.
 - **The two 3730 rows do not reconcile**. The stated CAGRs (2.5% units, 5% revenue) do not reproduce the printed series (1 to 3 units, USD 0.3M to 0.8M). Every other CE row reconciles exactly against its stated CAGR, so this looks like a hand-entered row. Small in absolute value but it should be corrected.
 - **Two rows for the 3500 / 3500xL / SeqStudio 8 Flex / SeqStudio 24 Flex family carry no visible end-use label** in the file as shared, because the model name is truncated. We have taken the first row as R&D/forensic and the second as diagnostic use, based on the consistent ordering of every other model pair in the sheet and on the relative volumes (15 versus 25 units in 2023). **This should be confirmed** — these two rows are the largest single component of CE volume, at 48 of 69 units in 2025.
 - **The CE model-level unit breakdown sums to 68 in 2025 against a CE total of 69**, a rounding difference. The 2023 column ties exactly at 52. Not material, but worth knowing the model-level rows are rounded.
 - **"Company A", "Company B" and "Company C" are unnamed** in the CE reagent breakdown. Companies B and C and "other manufacturers" are all zero, so the sheet assumes a three-supplier CE reagent market with no residual.
 - **Company A has reagent revenue but no software, parts or service revenue**, consistent with a reagent-only supplier that sells no instruments.
-

Appendix D — Transcript reliability and corrections applied

The transcript is a machine transcription with no speaker labels. Two consequences: speaker attribution in this memo is inferred from context, and a number of terms were mis-heard. The corrections we applied:

As transcribed	Read as	Basis
"Sanofisher"	Thermo Fisher	Context — he is describing the employer he has left
"cDNA sequencer", "CAGR units", "CAC cancels", "skin sequencers"	CE (capillary electrophoresis) sequencer / CE units	Context throughout; the questionnaire subject is CE
"short rate"	short read	Standard term for NGS short-read sequencing
"assets"	assays	Context — discussing assay kits and TSO 500
"BNTR kits"	VNTR kits	Standard forensic term (variable number tandem repeat)
"identifier kits"	Identifiler kits	Thermo Fisher product name, also listed in the assay table
"aims"	AIIMS	Context — "19", "13 to 14 fully functional", medical institutions
"autoflex"	SeqStudio Flex (probable)	Context — the successor to the 3500 in the Thermo Fisher CE line. Should be confirmed
"MCC"	Maternal cell contamination (probable)	Standard usage alongside chimerism monitoring
"Jaitanya" / "Chaitanya" / "Chetan" / "Jatin"	Chaitanya and Jatin	Two people on our side; the transcript renders both names several ways
"Shailendra"	Sailesh (probable)	Appears where the data-sheet author is referenced

Coverage limitation. The transcription ends at 30 minutes with "This file is longer than 30 minutes. Go Unlimited at TurboScribe to transcribe files up to 10 hours long." The final usable line is Devapriya saying "Chetan, do we have any other points to confirm? Maybe regarding the —", cut off mid-sentence. The five unanswered questions in Appendix A all fall after this point. **The full recording should be re-transcribed before this memo is treated as complete**, because some or all of those five may already have been answered.

Appendix E — Key figures in one place (2025 unless stated)

Figure	Value	Source
Total DNA sequencer market, sheet lines only	USD 110.7M	Sheet
Total DNA sequencer market, his verbal estimate	~USD 120M	Call
Total DNA sequencer market, adjusted for both excluded assay pools	USD 126–134M	Our calculation
CE equipment revenue	USD 5.4M	Sheet
CE reagent revenue (running reagents only, excludes assays)	USD 8.2M	Sheet, confirmed on call
CE diagnostic assay kit revenue	USD 4.5M	Sheet, corrected reading
CE forensics assay revenue (excluded from the model)	USD 5–8M	Sheet note
CE software and maintenance	USD 1.2M	Sheet
CE units sold	69	Sheet
CE implied average price per unit	~USD 78,000	Our calculation
NGS short-read equipment revenue	USD 24.2M	Sheet
NGS short-read reagent revenue (includes OEM assays)	USD 57.6M	Sheet
NGS short-read units sold	79	Sheet
NGS short-read implied average price per unit	~USD 306,000	Our calculation
NGS long-read equipment revenue	USD 4.7M	Sheet

Figure	Value	Source
Third-party NGS assays (excluded from the model)	USD 10-15M	Call
Thermo Fisher share of CE instruments	~95% (94.1% units, 94.6% revenue)	Call, verified against sheet
Thermo Fisher share of CE reagents	71%	Sheet
CE-relevant genetic tests performed in India	~500K-600K per year	Follow-up questionnaire data
Government share of clinical-diagnostics CE units	~one third (8 of 24)	Call
CE equipment CAGR, 2023-2035	10.6% (printed as 9.76%)	Sheet, corrected
CE reagent CAGR, 2023-2035	8.0% (printed as 7.36%)	Sheet, corrected
CE units in 2035 as modelled / with his 1-2 pp cut	174 / 156-140	Sheet and our calculation

Appendix F — Source data extract

The Excel model as shared, extracted from the PDF versions, so every figure quoted in the memo can be checked. All revenue figures are **USD million** unless stated otherwise.

Files received (several were duplicates of one another): the main data sheet appeared as data-1, data-2, DNA_Sequencer_Market_Information_Collection_Sailesh_updated_Analysis and ...Analysis (1); the CAGR sheet appeared as data-3 and ...Analysis (2). Content is identical except that the Analysis versions carry the full model name for the 3500 / SeqStudio Flex rows, which is truncated in the data-1 / data-2 renderings.

The **CAGR column** in the source is computed over 13 years rather than 12 intervals and is therefore understated by roughly 0.8 percentage points throughout. The "actual" column below is our correction.

Table 1 — Number of units sold

Line	Price range	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR in sheet
Total DNA sequencers	-	137	153	171	187	205	224	246	269	295	324	356	390	429	-
NGS short read (all mfrs, all models)	40K-800K	65	72	79	87	95	105	115	127	139	153	169	185	204	10
NGS long read (all mfrs, all models)	20K-300K	20	22	23	25	27	29	32	34	37	40	43	47	50	8
CE (all mfrs, all models)	-	52	60	69	75	82	90	99	108	119	131	144	158	174	-

CE breakdown by manufacturer, model and end use — units

Manufacturer	Model	Price range	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR	Note in sheet
Thermo Fisher	3730 (96-capillary, large) — R&D / forensic identification	250-300K	1	1	2	2	2	2	2	3	3	3	3	3	3	2.5	
Thermo Fisher	3730 (96-capillary, large) — diagnostic use	250-300K	0	0	1	1	1	1	1	1	1	1	1	1	2	2.5-5	
Thermo Fisher	3500, 3500xL, SeqStudio 8 Flex, SeqStudio 24 Flex (8-24 capillary) — R&D / forensic (inferred)	75-150K	15	16	17	18	20	21	23	24	26	28	30	32	34	7	
Thermo Fisher	3500, 3500xL, SeqStudio 8 Flex, SeqStudio 24 Flex (8-24 capillary) — diagnostic use (inferred)	75-150K	25	28	31	35	39	44	49	55	62	69	78	87	97	12	
Thermo Fisher	3500 Dx, 3500xL Dx, SeqStudio 8 Flex Dx, SeqStudio 24 Flex Dx	75-150K	2	2	2	2	2	3	3	3	3	3	3	3	4	5	Non-regulated market
Promega	Spectrum CE system (mid-range) — R&D / forensics	100-120K	0	1	1	1	1	1	1	1	1	2	2	2	2	5	Only fragment - no marketing
Promega	Spectrum CE system (mid-range) — diagnostic use	100-120K	0	0	0	0	0	0	0	0	0	0	0	0	0	0	No Sanger seq, only fragment
Thermo Fisher	SeqStudio (4-capillary, small) — R&D / forensic	50-60K	0	1	1	1	1	1	1	1	1	2	2	2	2	5	Not promoted by TMO
Thermo Fisher	SeqStudio (4-capillary, small) — diagnostic use	50-60K	8	9	10	11	12	13	14	16	17	19	21	23	25	10	Only standalone lab / tier-2, tier-3 labs
Promega	Spectrum Compact (4-capillary, small) — R&D / forensics	50-70K	0	1	1	1	1	1	1	1	1	2	2	2	2	5	Unless Promega focus on aggressive marketing
Promega	Spectrum Compact (4-channel, compact) — diagnostic use	50-70K	1	1	2	2	2	3	3	3	3	4	4	4	4	8	Unless Promega focus on aggressive marketing
Other manufacturers	-	-															(blank)

The end-use label for the two 3500 / SeqStudio Flex rows is truncated in every file we received. We have taken the first as R&D / forensic and the second as diagnostic use, on the basis that every other model pair in the sheet is ordered that way and that diagnostic volumes are the larger of the two everywhere else. **This needs confirming** — together these two rows are 48 of 69 CE units in 2025.

Cross-checks: the CE model rows sum to 52 in 2023 (ties exactly to the CE total) and 68 in 2025 (against a CE total of 69 — rounding). Thermo Fisher accounts for 64 of 68 units in 2025, or 94.1%, consistent with his verbal "95 percent".

Table 2 — Equipment sales revenue

Line	Price range	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR in sheet	Actual CAGR
Total DNA sequencers	-	27.9	30.8	34.3	37.5	41.1	45.1	49.4	54.2	59.4	65.1	71.4	78.3	85.9	9.04%	9.83%
NGS short read	40K-800K	20.0	22.0	24.2	26.6	29.3	32.2	35.4	39.0	42.9	47.2	51.9	57.1	62.8	9.20%	10.00%
NGS long read	20K-300K	4.0	4.3	4.7	5.0	5.4	5.9	6.3	6.9	7.4	8.0	8.6	9.3	10.1	7.36%	8.00%
CE	-	3.9	4.5	5.4	5.9	6.4	7.0	7.6	8.3	9.1	10.0	10.9	11.9	13.1	9.76%	10.62%

CE breakdown — equipment revenue

Manufacturer	Model / use	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR
Thermo Fisher	3730 — R&D / forensic	0.3	0.3	0.5	0.5	0.6	0.6	0.6	0.6	0.7	0.7	0.7	0.8	0.8	5
Thermo Fisher	3730 — diagnostic	0.0	0.0	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	2.5-5
Thermo Fisher	3500 / SeqStudio Flex family — R&D / forensic	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.1	2.2	2.4	2.5	2.7	7
Thermo Fisher	3500 / SeqStudio Flex family — diagnostic	1.8	2.0	2.3	2.5	2.8	3.2	3.6	4.0	4.5	5.0	5.6	6.3	7.0	12
Thermo Fisher	3500 Dx / SeqStudio Flex Dx family	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4	5
Promega	Spectrum CE — R&D / forensics	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	5
Promega	Spectrum CE — diagnostic	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
Thermo Fisher	SeqStudio small — R&D / forensic	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	5
Thermo Fisher	SeqStudio small — diagnostic	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.8	0.9	0.9	1.0	1.1	1.3	10
Promega	Spectrum Compact — R&D / forensics	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	5
Promega	Spectrum Compact — diagnostic	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	8

For ten of these eleven rows the revenue CAGR equals the unit CAGR, meaning **the model holds average selling prices flat across the whole forecast**. The only exception is the 3730 R&D row, at 2.5% units against 5% revenue.

Table 3 — Reagent sales

Line	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR in sheet	Actual CAGR	Note in sheet
Total DNA sequencers	50.0	58.9	69.4	81.9	96.9	114.6	135.9	161.2	191.4	227.5	270.6	322.1	383.7	16.97%	18.51%	
NGS short read	40.0	48.0	57.6	69.1	82.9	99.5	119.4	143.3	172.0	206.4	247.7	297.2	356.6	18.33%	20.00%	Includes OEM assays
NGS long read	3.0	3.3	3.6	4.0	4.4	4.8	5.3	5.8	6.4	7.1	7.8	8.6	9.4	9.20%	10.00%	Includes OEM assays
CE	7.0	7.6	8.2	8.8	9.5	10.3	11.1	12.0	13.0	14.0	15.1	16.3	17.6	7.36%	8.00%	Forensics assays — 5-8 M not included

CE breakdown — reagent sales by supplier

Supplier	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	Share 2023
Thermo Fisher Scientific	5.00	5.4	5.8	6.3	6.8	7.3	7.9	8.6	9.3	10.0	10.8	11.7	12.6	71.4%
Promega	1.00	1.08	1.2	1.3	1.4	1.5	1.6	1.7	1.9	2.0	2.2	2.3	2.5	14.3%
Company A (unnamed; Nimagen per our open question)	1.00	1.08	1.2	1.3	1.4	1.5	1.6	1.7	1.9	2.0	2.2	2.3	2.5	14.3%
Company B	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0%
Company C	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0%
Other manufacturers	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0%

Promega and Company A are modelled as exactly equal, at the same growth rate, for all thirteen years. The three named suppliers sum exactly to the CE total, so the sheet assumes a three-supplier CE reagent market with no residual.

Definition confirmed on the call: the CE reagent line is running reagents only and **excludes assay kits**. The NGS reagent lines **include** OEM assays plus running cost. This asymmetry is the main definitional problem in the model.

Table 4 — Software, parts and other sales

Line	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	CAGR in sheet	Actual CAGR	Note in sheet
Total DNA sequencers	5.5	6.2	7.1	8.0	9.1	10.3	11.7	13.3	15.2	17.2	19.6	22.4	25.5	12.52%	13.63%	
NGS short read	4.0	4.6	5.3	6.1	7.0	8.0	9.3	10.6	12.2	14.1	16.2	18.6	21.4	13.77%	15.00%	Only the maintenance for NGS; softwares FOC from OEM; 3rd party reporting softwares not included
NGS long read	0.5	0.55	0.6	0.7	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.6	9.20%	10.00%	Same note as above
CE	1.0	1.08	1.17	1.26	1.36	1.47	1.59	1.71	1.85	2.00	2.16	2.33	2.52	7.36%	8.00%	Software and maintenance included
— of which Thermo Fisher	0.95	1.03	1.11	1.20	1.29	1.40	1.51	1.63	1.76	1.90	2.05	2.22	2.39	7.36%	8.00%	
— of which Promega	0.05	0.05	0.06	0.06	0.07	0.07	0.08	0.09	0.09	0.10	0.11	0.12	0.13	7.36%	8.00%	
— Company A / B / C, other	(blank)															

NGS short read rises from 72.7% of this line in 2023 to 84.0% in 2035, entirely because it is assumed to grow at 15% against CE at 8% and NGS long read at 10%. Company A has reagent revenue but nothing here, consistent with a reagent-only supplier.

Table 5 — Diagnostic assay reagents using CE

Header in the source: "Diagnostic assay reagents using CE (refers to assay reagent kits specialized for specific diagnostics, excluding electrophoresis reagents (polymer and buffer), STR reagents for personal identification, and fluorescent sequencing reagents)"

The column below is headed "Annual Kit Sales Volume – Million \$" in the source. **Sailesh confirmed on the call that it is annual kit sales *revenue* in USD million, not volume**, and that the header is wrong: "I should have mentioned revenues in itself directly."

Manufacturer	Product name	Approval status (Indian medical device authorities)	Target disease	Target gene	Annual kit sales revenue, USD M	Price range, USD
Thermo Fisher	MSI	(blank)	Cancer	Multiple MS	0.1	25
Thermo Fisher	FGA+SMA	(blank)	Fragile X and SMA	Fragile X and SMA	0.1	25
Thermo Fisher	Identifiler	(blank)	Chimerism, MCC	Multiple STR	0.8	20
Promega	Forensic kit	(blank)	Chimerism, MCC	Multiple STR	0.8	20
Promega	MSI	(blank)	Cancer	Multiple MS	0.2	25
MRC Holland	MLPA	(blank)	Multiple diseases	Multiple loci	1.0	25
Asuragen	FGA+SMA	(blank)	FGA+SMA	FGA+SMA	0.5	25
Devyser	QF-PCR	(blank)	Pre-natal testing	Multiple loci	1.0	25
Total					4.5	

Reading the column as volume and multiplying by the price range gives **USD 104.5M**, which is the figure computed live on the call and then corrected. The approval status column is blank for every product. Identifiler and the Promega Forensic kit are both multiplex STR products, which the table header says are excluded, yet both are recorded against clinical applications (chimerism monitoring and maternal cell contamination).

Table 6 — Tests performed, from the second follow-up questionnaire

Supplied by Sailesh before the call. The "tests originated" column was what we asked him to fill in and is still outstanding.

Provider type	Tests originated	Tests performed
Specialised genomics labs	(requested)	~40% (~200K-240K)
National chain labs	(requested)	~20% (~100K-120K)
Regional chain labs	(requested)	Negligible
Standalone labs	(requested)	Negligible
Specialty treatment hospitals	(requested)	~10% (~50K-60K)
Large private hospitals	(requested)	~10-15% (~50K-90K)
Small and medium private hospitals	(requested)	Negligible
Government tertiary hospitals / public medical colleges / programme-linked centres	(requested)	~15-20% (~75K-120K)

Note in the source: total CE-relevant genetic tests done in India ~500K-600K.

Cross-check: the shares sum to 95-105% and the absolute volumes to 475K-630K, against a stated total of 500K-600K. Internally consistent.

Derived figures used in the memo

All computed from the tables above.

Figure	Value	How derived
Total market 2025, sheet lines only	USD 110.7M	34.3 equipment + 69.4 reagents + 7.1 software
Total market 2025, adjusted	USD 126-134M	110.7 + 10-15 third-party NGS assays + 5-8 CE forensics assays
CE implied price per unit, 2025	~USD 78,000	5.3963M ÷ 69 units
NGS short-read implied price per unit, 2025	~USD 306,000	24.2M ÷ 79 units
NGS long-read implied price per unit, 2025	~USD 203,000	4.6656M ÷ 23 units
CE reagent revenue per unit shipped, 2023 → 2035	USD 134,600 → USD 101,300 (-25%)	CE reagents ÷ CE units, each year
CE end-use split 2025	22 R&D/forensic, 46 diagnostic	Sum of the model rows by end use
CE end-use split 2023	16 R&D/forensic, 36 diagnostic	Sum of the model rows by end use
Thermo Fisher share of CE units, 2025	94.1%	64 of 68 model-level units
Thermo Fisher share of CE equipment revenue, 2025	94.6%	5.3M of 5.6M
CE units 2035 with a 1 pp CAGR cut	156	52 × 1.0959 ¹²
CE units 2035 with a 2 pp CAGR cut	140	52 × 1.0859 ¹²

Figure	Value	How derived
CE revenue mix 2025 incl. diagnostic assays	28.1% equipment / 65.9% reagents+assays / 6.1% software	$(5.40, 8.16+4.5, 1.17) \div 19.23$
NGS software/service gap, verbal vs sheet	+USD 4.1M	10.0 verbal – 5.9 sheet
NGS reagents+assays gap, verbal vs sheet	+USD 23.8M to +28.8M	85-90 verbal – 61.2 sheet